Leina Kameron

MSc Bioinformatics Candidate · Computational Biology & RNA-seq Analysis

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PROFILE

Bioscience graduate with an MSc in Bioinformatics, specialised in transcriptomic data analysis, RNA-seq pipelines, and co-expression network analysis. Built a fully functional standalone research software platform (ARCANA) integrating DESeq2, WGCNA, and machine learning to study lncRNA networks in pancreatic islets in Type 2 Diabetes. Comfortable working independently at the interface of wet-lab biology and computational analysis, with a rigorous approach to data quality, reproducibility, and publication-ready output.

EDUCATION

MSc Bioinformatics

2024 – Present

Birkbeck, University of London

Dissertation: ARCANA — lncRNA Co-expression Network Analysis Platform for Pancreatic Islet T2D Research (GSE164416)

BSc Bioscience

2021 – 2024

Manchester Metropolitan University

Foundations in molecular biology, cell biology, genetics, biochemistry, and laboratory techniques

RESEARCH PROJECT

ARCANA — lncRNA Co-expression Analysis Platform

Python · R · JavaScript · FastAPI · XGBoost

MSc Dissertation · Birkbeck University · 2024–Present · Dataset: GSE164416 (36 pancreatic islet donors; T2D, IGT, ND)

- Designed and built a full end-to-end RNA-seq analysis pipeline handling quality control, normalisation, differential expression (DESeq2, apeglm shrinkage), and co-expression network construction (WGCNA) — producing **30 gene modules** and identifying **604 significant differentially expressed genes** across T2D vs ND, IGT vs ND, and T2D vs IGT comparisons.
- Ranked hub lncRNAs in disease-associated modules using intramodular connectivity, eigengene correlation, and an **XGBoost machine learning classifier** (5-fold cross-validation) with SHAP explainability to derive per-gene importance scores.
- Engineered a **standalone web application** (FastAPI backend + live dashboard frontend) with WebSocket progress streaming, SQLite run-history database, full-text document search, and a REST API — making the full analysis pipeline operable without the command line.
- Generated 19 publication-quality figures (300 dpi PNG / PDF / SVG) with auto-written methods text, figure legends, and supplementary hub gene cards.
- Resolved complex environment compatibility issues across Python, R, and macOS to achieve a reproducible, portable installation (conda environment + requirements file).

DESeq2 · WGCNA · XGBoost · SHAP · RNA-seq · lncRNA · FastAPI · SQLite · pandas · ggplot2 · Bioconductor · matplotlib · seaborn

TECHNICAL SKILLS

Languages	Python (advanced), R (advanced), JavaScript, Bash / Shell scripting
Bioinformatics	RNA-seq pipeline design, DESeq2, WGCNA, Bioconductor, gene annotation (Ensembl / BioMart), lncRNA classification, bulk transcriptomics
Machine Learning	XGBoost, scikit-learn, SHAP explainability, cross-validation, feature ranking
Data Analysis	pandas, numpy, scipy, statistical hypothesis testing, PCA, UMAP, clustering
Visualisation	ggplot2, matplotlib, seaborn, publication-ready 300 dpi figure generation
Software / Tools	FastAPI, SQLite, conda / pip, Git, WebSocket, REST APIs, uvicorn, Linux / macOS CLI
Biological Knowledge	Transcriptomics, lncRNA biology, pancreatic islet physiology, Type 2 Diabetes molecular mechanisms, gene regulation

RELEVANT COURSEWORK

Genomics & Transcriptomics · Statistical Bioinformatics · Machine Learning for Biosciences · Molecular Biology & Genetics · Cell Biology · Biochemistry · Database Systems · Scientific Programming

KEY COMPETENCIES

- Experienced in handling large, multi-sample genomic datasets with rigorous QC, normalisation, and reproducible analytical workflows
- Strong scientific writing: can produce methods sections, results summaries, and figure legends to publication standard
- Independent problem-solver: debugged complex cross-platform dependency conflicts across Python, R, and macOS system libraries
- Able to communicate technical results clearly in written and visual form for both specialist and non-specialist audiences